

Bayesian Spatial Search — Deliverable 2

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1 Prior distribution

The prior $\pi_j = \mathbb{P}(Z = j)$ is built from the physics of the accident and the two witness statements:

$$Z = x_E + \alpha v_{\text{plane}} + \tau (v_{\text{wind}} + v_{\text{drift}}) + 1.5\varepsilon + \eta \hat{n}_{\perp}.$$

Justification. The error term $\varepsilon \sim \mathcal{N}(0, I_2)$, stands for a general 2D uncertainty (it can go in both directions). I_2 is the identity matrix and multiplying it by 1.5 we get $\varepsilon \sim \mathcal{N}(0, \text{diag}(2.25, 2.25))$. The $\hat{n}_{\perp}$ term is the unit vector orthogonal to v_{plane} , hence $\eta \hat{n}_{\perp}$ with $\eta \sim \mathcal{N}(0, 1)$ simulates a lateral error term. The perpendicular term $\sigma_{\perp} \eta$ encodes Witness 2’s qualifier “*slightly off to one side*”: a symmetric perpendicular spread.

Parameters. $\alpha = 1$ corresponds to the canonical horizontal displacement implied by v_{plane} . We set $\alpha \sim \mathcal{N}(1.5, 0.5^2)$ because Witness 2 reports that the object “*fell farther along the trajectory*” (mean > 1), and Witness 1’s report that it “*kept moving forward*” rules out values close to zero.

For the drift phase, τ has no direct witness information, so we use $\tau \sim \mathcal{N}(1.0, 0.4^2)$ centred on the canonical drift magnitude.

Since Z is a nonlinear function of α and τ (products of Gaussian scalars with deterministic vectors), the marginal density of Z is not Gaussian. We use Monte Carlo discretization to avoid having to compute or approximate this density analytically. We sample ($N = 2 \cdot 10^5$ draws), counting how many landing samples fall in each cell and normalize to obtain π_j with $\sum_j \pi_j = 1$. We discard any possible points that lie outside the grid (they are really unlikely).

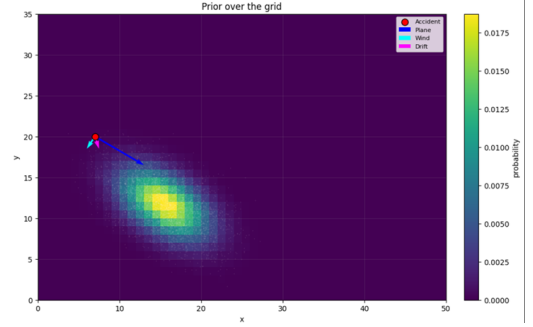


Figure 1: Prior spatial distribution.

2 Detection model

We model the conditional detection probability given coverage with a hazard function:

$$g(d_j, r_j) = \kappa \exp(-\gamma_d d_j - \gamma_r r_j),$$

where d_j and r_j are the depth and roughness respectively, at cell j . The associated penalties, $\gamma_d = 1.8$ and $\gamma_r = 1.5$ are set to these values after some calibration testing, and the fact $\gamma_r < \gamma_d$ is because the previous mission reports suggest that strong depths effects and rugged terrain severely reduce detection in this order of importance. It is a matter of choice, not any strong knowledge. Then the parameter k is the one that models how is effort going to affect the final detection probability.

We calculate the detection probability as:

$$\rho_{tj} = 1 - \exp(-e_t \cdot g(d_j, r_j)),$$

so that $q_{tj} = c_{tj} \rho_{tj}$ with $c_{tj} \in \{0, 1\}$. And so, in a perfect terrain ($d_j = d_r = 0$) with $k = 0.9$ we have probabilities 59%, 83%, 93% with respect to the three effort levels. We have consciously chosen an imperfect detection model: even in the perfect terrain with minimum effort we get 59% chance of finding the object. This is purely a design choice we have made.

3 Posterior update

Given a mission t over rectangle R_t with effort e_t and outcome $s_t \in \{0, 1\}$, the likelihood under the model is

$$\mathbb{P}(s_t | Z = j) = (c_{tj} \rho_{tj})^{s_t} (1 - c_{tj} \rho_{tj})^{1-s_t},$$

which, by Bayes' rule with normalising constant Z_t , gives the multiplicative update

$$\pi_j^{(t)} = \frac{1}{Z_t} \pi_j^{(t-1)} \cdot [s_t c_{tj} \rho_{tj} + (1 - s_t) (1 - c_{tj} \rho_{tj})].$$

After a non-detection ($s_t = 0$), cells outside R_t are unaffected ($c_{tj} = 0$ implies likelihood 1) and cells inside lose mass proportionally to their detectability. Crucially, cells inside R_t with high depth or roughness lose *less* mass than smooth shallow cells, because their ρ_{tj} is smaller: non-detection in hostile terrain is weak evidence, which matches intuition.

4 Mission strategy

With a budget of 530, a 12-hour cooldown and a tight deadline (I am writting this report 6 days before deadline), an effective plan must do at most 6-8 missions. So each mission must yield meaningful information. In the streamlit app, I have checked that the dollar cost is exactly the number of cells times the effort: $|R| \cdot e$. This is why we have set a **minimum number** of cells per mission: $|R| \geq m$. There is no point in small missions if the cost is proportional to the covered area and there is a large cooldown time.

We score candidate missions by their efficiency

$$\text{Eff}(R, e) = \frac{\mathbb{E}[s_t = 1]}{\text{cost}} = \frac{\sum_{j \in R} \pi_j \rho_{tj}(e)}{|R| \cdot e},$$

and implement a brute-force search over the space of all possible rectangles with a set of constraints:

1. $|R| \geq m$
2. cost > budget remaining
3. `minimum_prob_detection` $\geq p$

The last part of the strategy lies in how do we choose parameters. For the two first missions we set a high m ($m = 50$) and a low p ($p = 0.2$) to enforce **exploration** rather than explotation. After the two first missions, if not found, the idea is to have a more concentrated posterior probability mass and enforce **explotation** with lower m and higher p .

5 Results

The first mission is rectangle $[10, 20] \times [9, 15]$ with effort 2 and the bomb was **not found**. The second mission is rectangle $[12, 21] \times [8, 14]$ with effort 3 and the bomb **was found**.

We can see in the image how the posteriors after both missions.

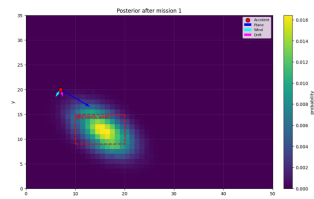


Figure 2: After Mission 1

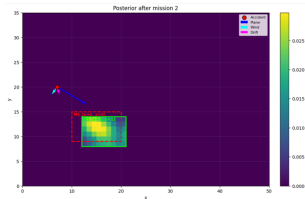


Figure 3: After Mission 2